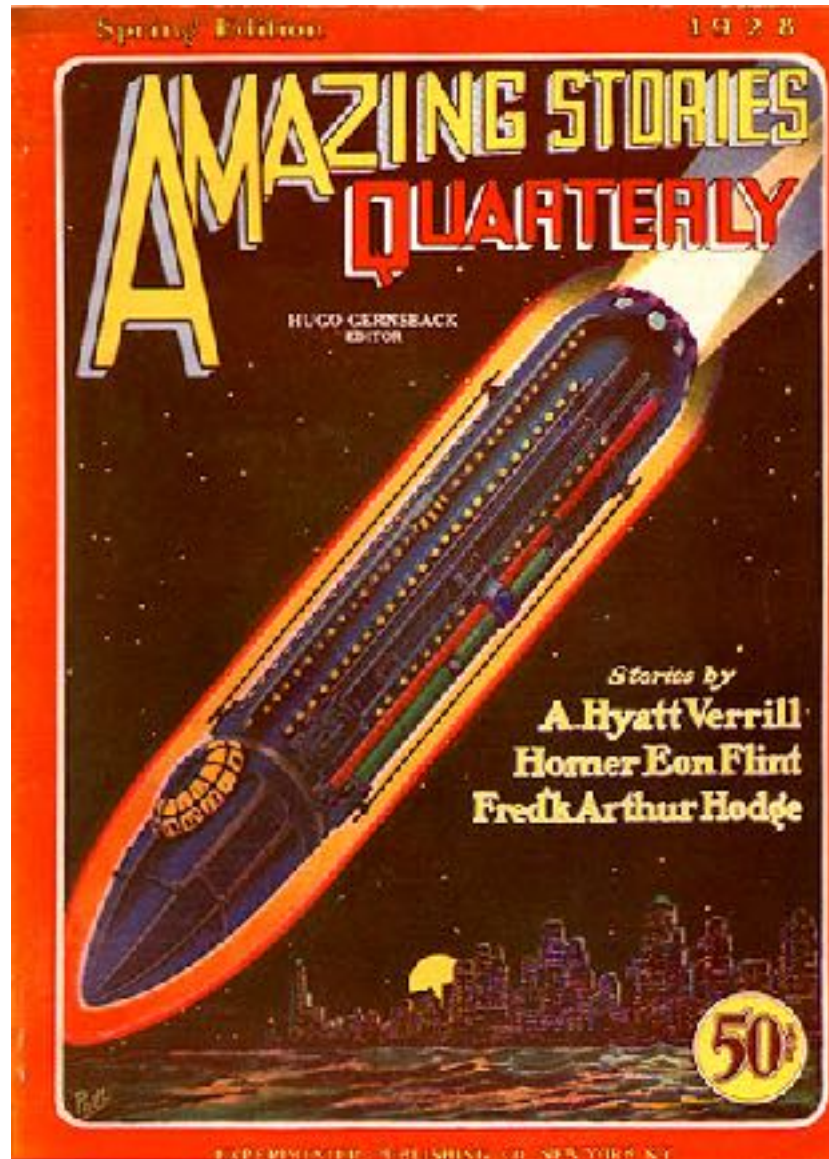




History of Science

Lecture 1: Introduction

History of Science

Overview:

- Overview of the course
- Prehistory of computing
 - Machines and technology
 - Three traditions
- A prehistory of AI
 - Human thinking and behaviour
 - Language!

History of Science

Overview of the course

- Martin Campbell-Kelly a.o., *Computer: a history of the information machine*, Westview (2023⁴)
- Approximately 40 hours of reading
- 14 hours lectures
- 9 hours (6 times 1,5h) working on assignments
- Canvas: extra info and quizzes
- Workgroups (enrol if you want to join)
- Which (in a 3EC course) leaves 20 hours for exam preparation

History of Science

Overview of the course: lectures

See the Canvas site (Modules) to be informed about which chapters to prepare.

During the lectures emphasis will be on the European side of the story and history of AI. American story in Campbell-Kelly. References to the book will be made occasionally.

Make notes while reading and during the lectures.

If during reading or lectures, you come across things you don't understand, please let me know!

History of Science

Overview of the course: assignments

Assignments are made in groups. Write essays!

Use details from the book and the lectures; if relevant from your own experience.

Team up and reflect on the text of your assignment.
Hand in a text that you can all relate to.

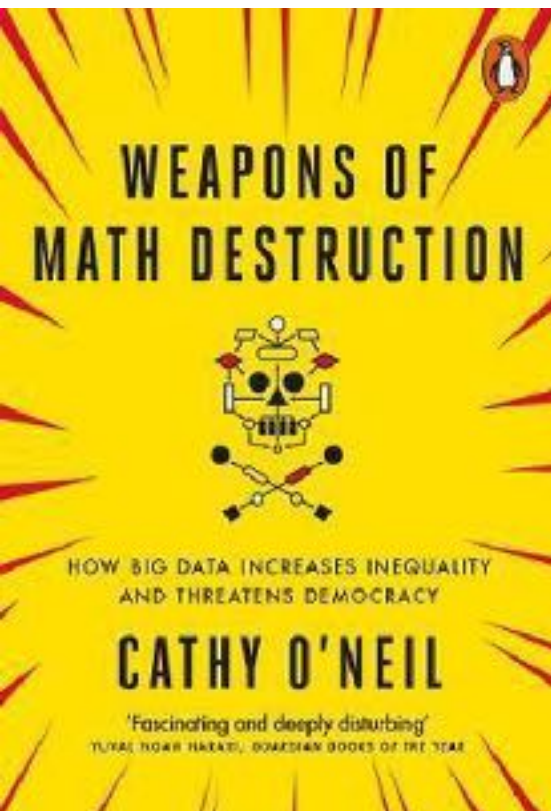
Assignments will be graded. You'll receive feedback.
If average of best four assignments is 6 or more,
you can skip one (designated) question on the
exam.

History of Science

Why this course?

Algorithms determine our lives. Digital culture.

Improvement? Definitely not in all cases. Change?



History of Science

Why this course?

Ok. But why history?

1. A history - not THE history
2. To tell a story about how this emerged
3. To tell a story about what it was that emerged
(digital culture? cf Haigh & Ceruzzi, 2021;
David Gugerli, 2022)

Reality, after all, has a way of “taking over”



History of Science

Why this course?

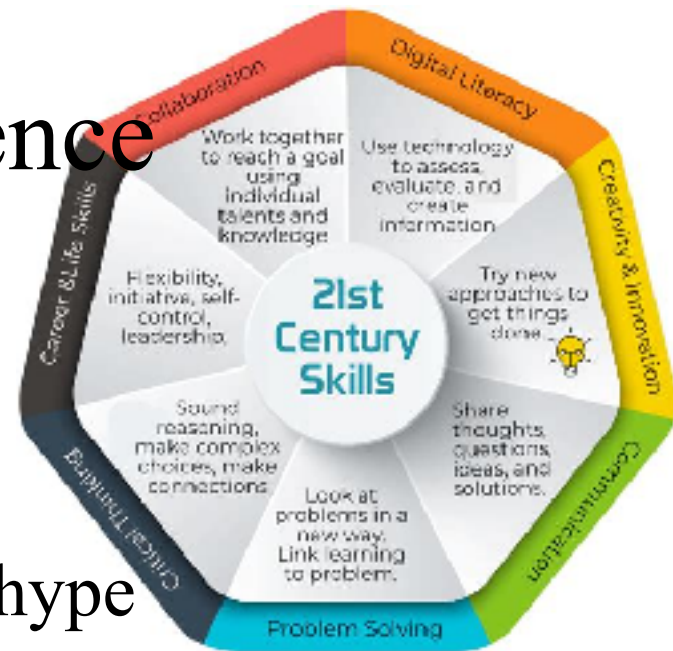
Ok. But why history?

To distinguish between reality and hype

To be confronted with your own prejudices.

When, for the first time in history, did a substantial number of people believe the earth was flat.

- A. 300 BC
- B. 1000 AD
- C. 1215 AD
- D. 2007 AD



History of Science

Why this course?

Out of comfort zone! Reflection on what we are, why we are (where we came from), where are we heading (and do you want to take another turn)?

NOT PRESCRIPTIVE: REFLECTIVE

- Historical: Campbell-Kelly
- Philosophical / ethical
- Sociological

History of Science

Why this course?

- Historical
 - Distance in time makes it easier to recognise several actors and interests
 - Distance in time makes choices from the past more visible
 - Historical failures won't help to prevent humankind making new ones, but...

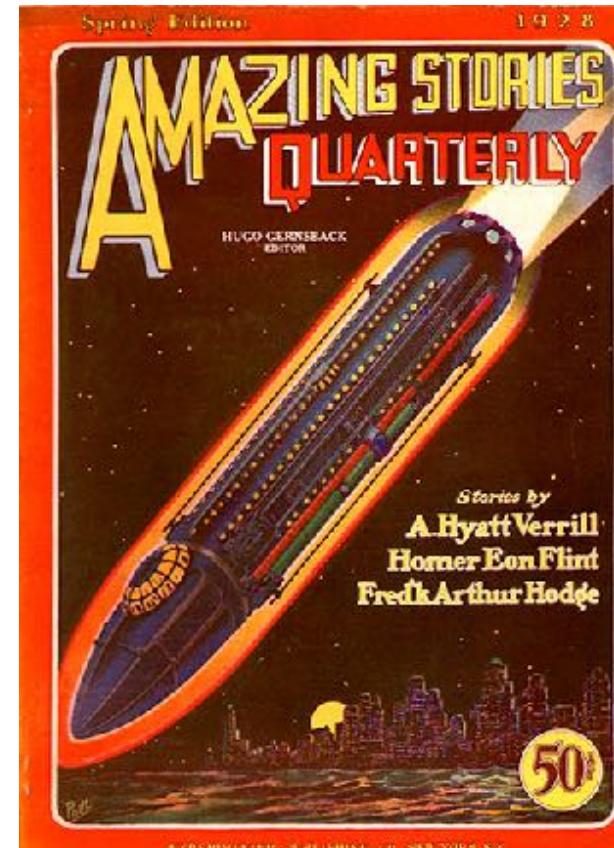
History of Science

Prehistory of computing

Roughly: Campbell-Kelly, chapters 1-3

Ideals made machines into a succes!

- Machines and technology
- Three traditions



History of Science

Machines and technology



History of Science

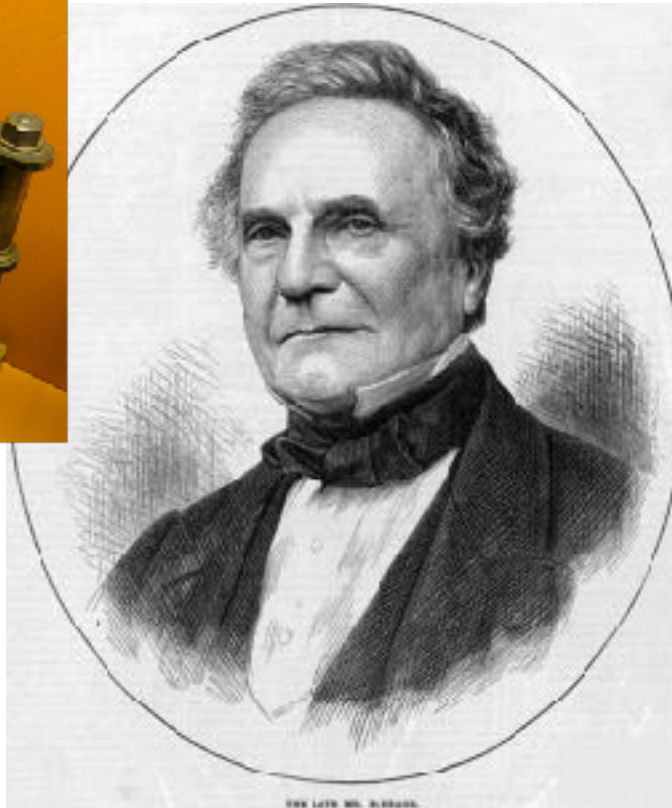
Machines and technology



History of Science

Charles Babbage (1791-1871)

Differential engine / analytical engine



History of Science

Charles Babbage

On the economy of machines and manufactures



W. H. & A. CO. LONDON

History of Science

Intermezzo

Charles Babbage revisited! Howard Aiken



History of Science

Frederik Taylor (1856-1915)

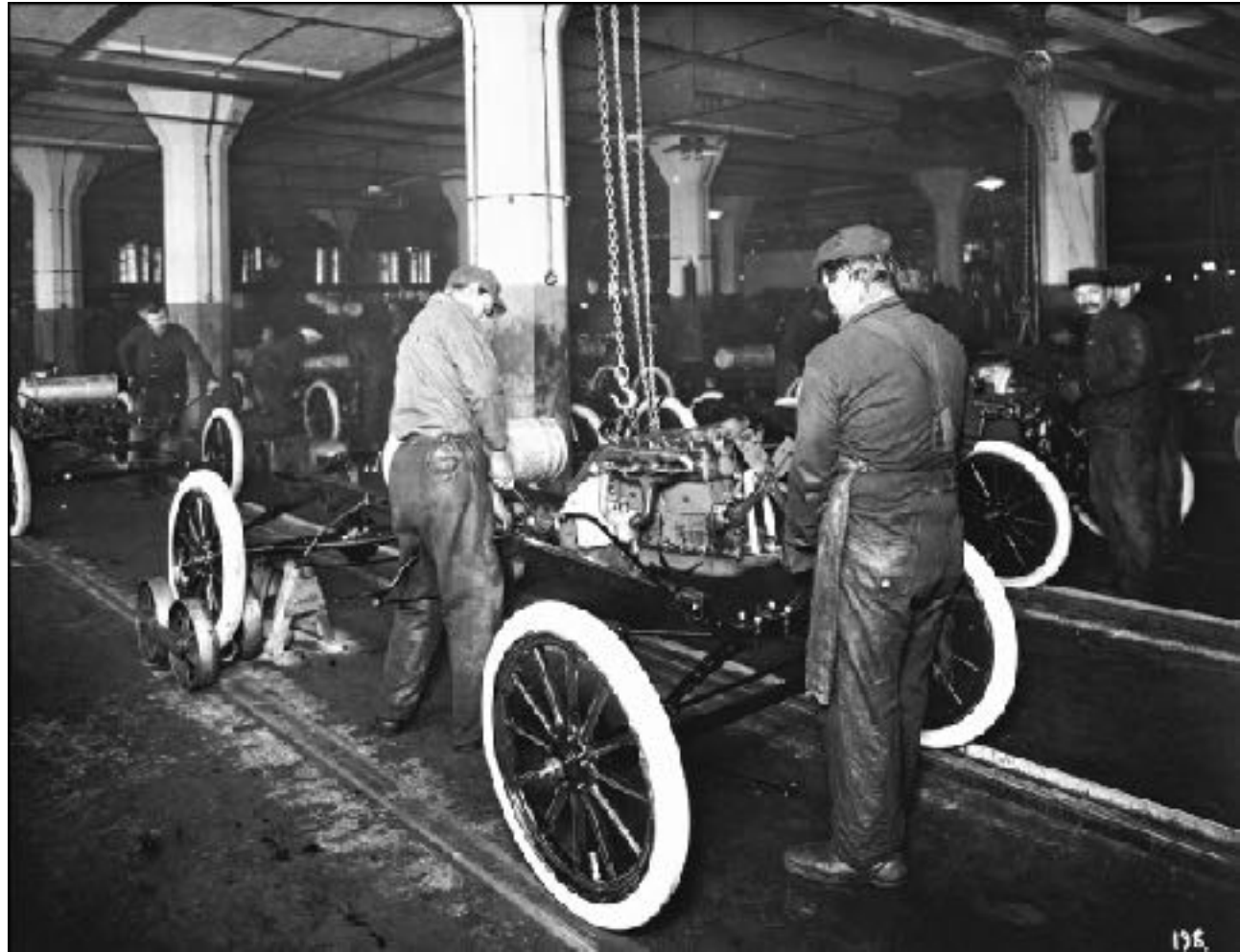
- *Shop management* (1903)
- *The principles of Scientific management* (1911)



History of Science

Frederik Taylor

Ford (1913)



History of Science

Frederik Taylor

In France during WW I.

Henri Fayol (1841-1925)

*Administration Industrielle et
générale* (1916)



History of Science

Three traditions or: three stories of automation



Administration



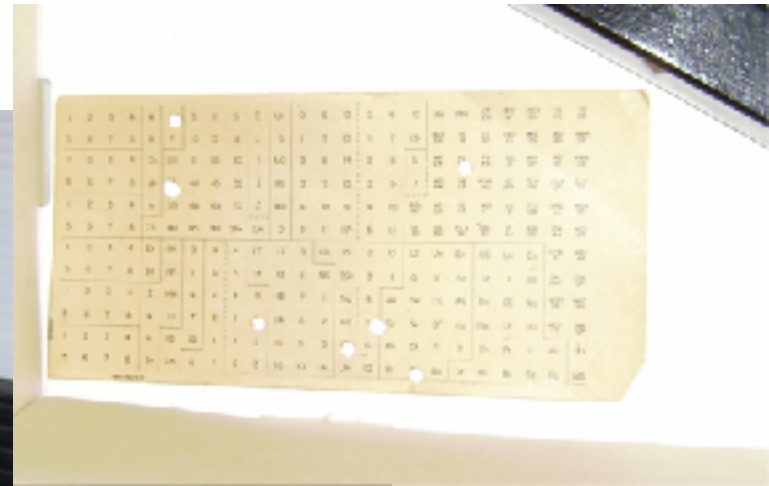
Process control



Science

History of Science

Administration: the 1890 US census
trust in Hollerith machines



History of Science

Three traditions (1): administration

The office in 1876



History of Science

Three traditions (1): administration

The office in 1920



History of Science

Three traditions (1): administration



Calculators (Odhner)

Typewriters (NCR, Olivetti, Odhner)



Punched card typing

Sorting machines

Counting machines

Tabulating machines (**IBM**,
Burroughs, ...)



History of Science

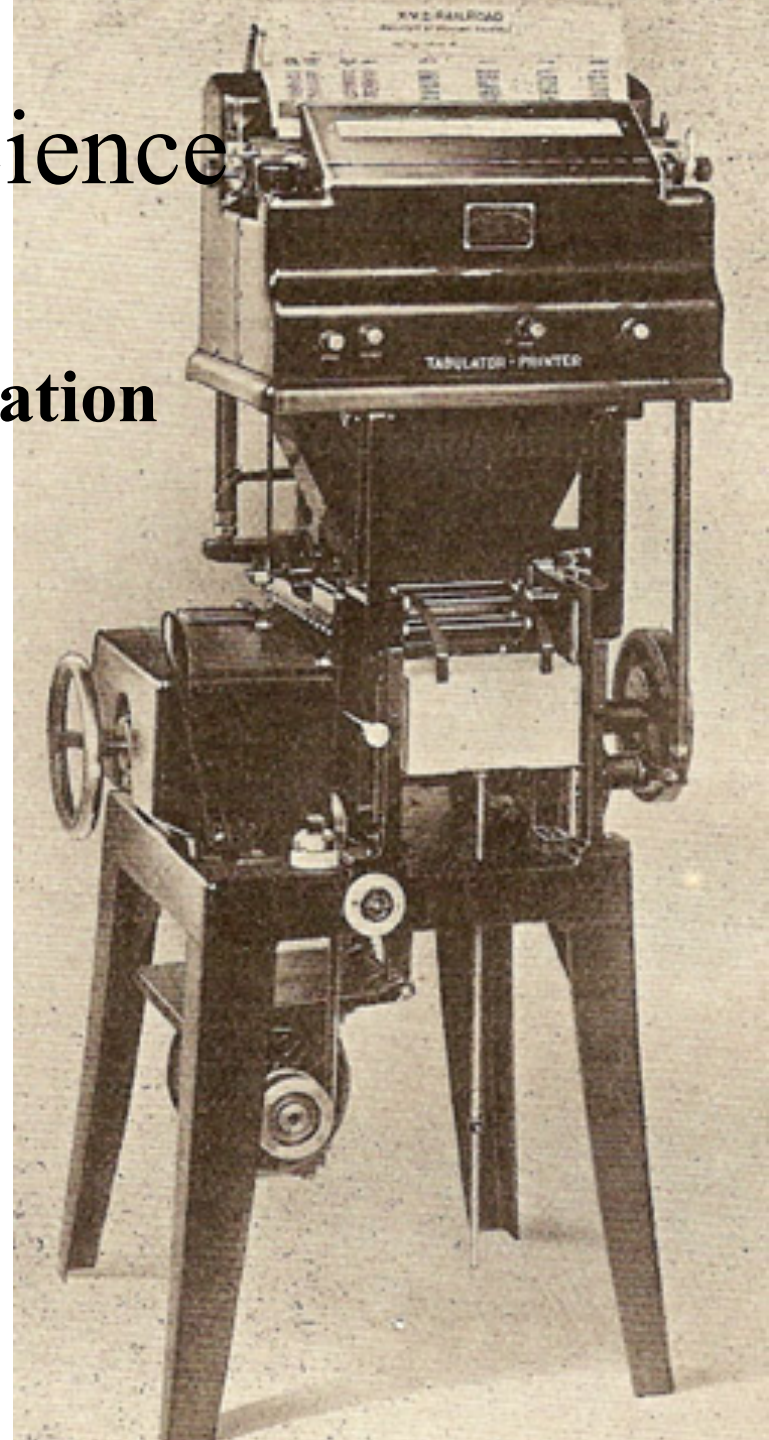
Three traditions (1): administration

Efficiency movement: journals,
institutions, 1911 Amsterdam:
*International Exhibition of Modern
Office Equipment*

ROBAVER 1922: success

PCGD 1923: failure

Dutch State Mines 1923: success



History of Science

Three traditions (1): administration

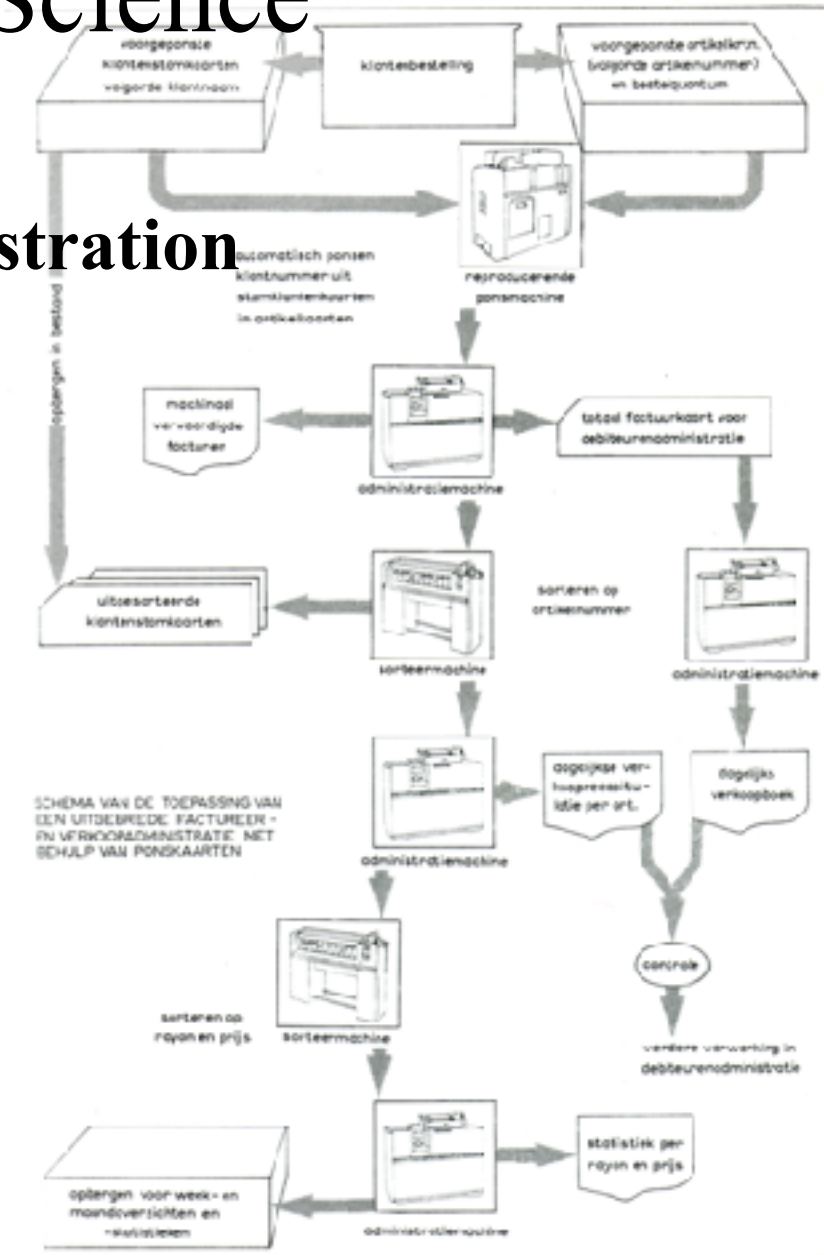
Not necessarily all very high tech.



History of Science

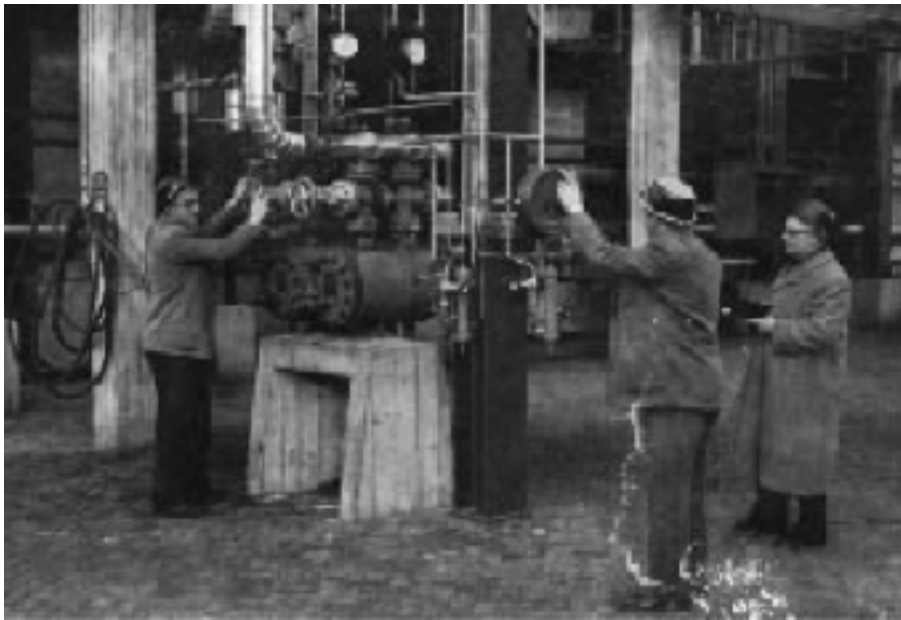
Three traditions (1): administration

But following a business outline

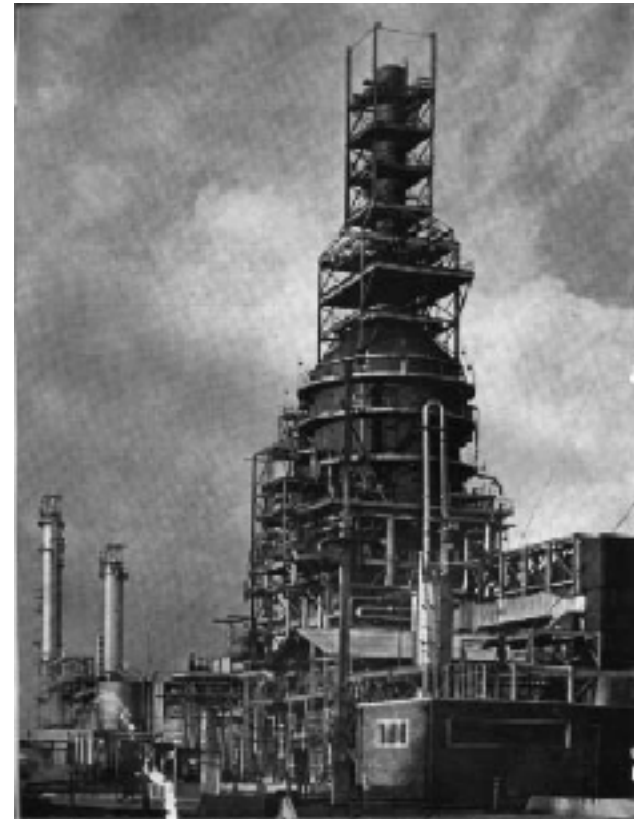


History of Science

Three traditions (2): process control



Shell (1935) / Hoogovens

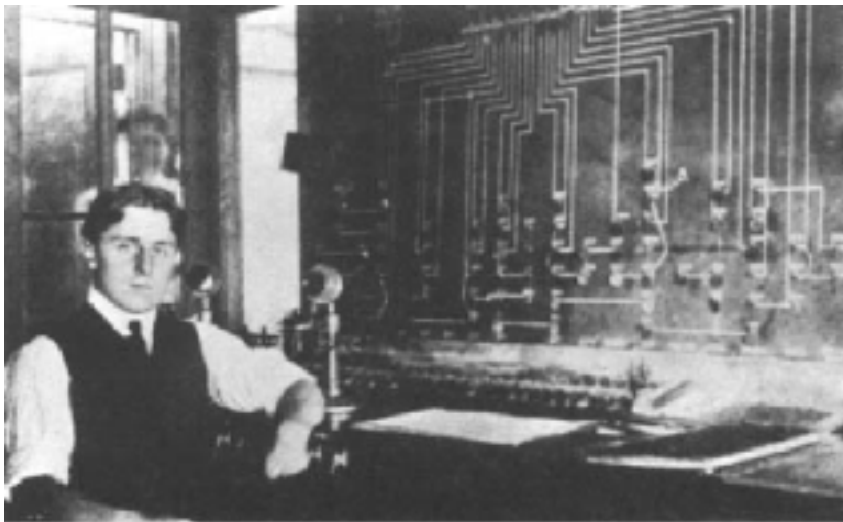


History of Science

Three traditions (2): process control

Outlines analogue to the process

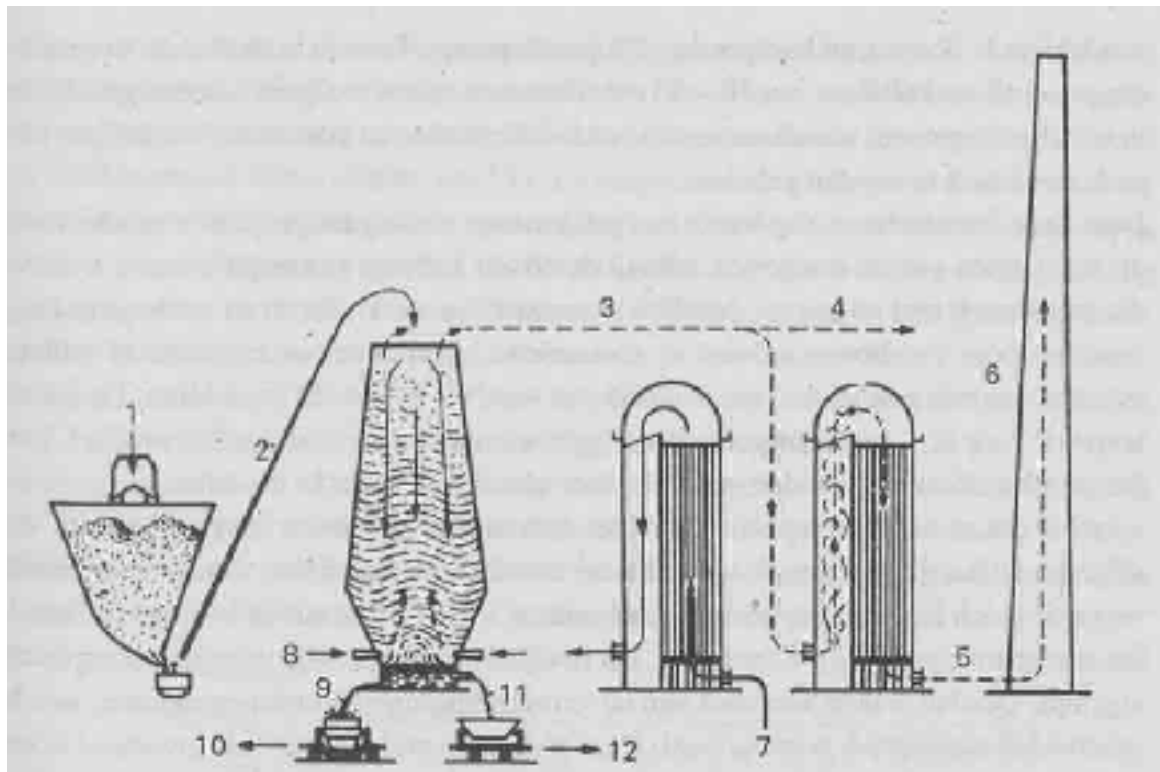
Edison (1898) or DSM (1956)



History of Science

Three traditions (2): process control

Outline of process (example from Hoogovens, 1930s)



History of Science

Three traditions (3): science and engineering

Computers (PEOPLE!) active in:

Weather (storm) prediction calculations

(Hydro)mechanical calculations

Aeronautics

Econometry (Jan Tinbergen)

Telephone networks

Military applications (tables for aiming large guns;
Manhattan project)



History of Science

Three traditions (3): science and engineering

D. Hartree (1897-1958), *Numerical Analysis*

V. Bush, Differential analyser (1934)



History of Science

Three traditions (3): science and engineering



Anhang 1a. Rechenformulare zum Houghschen Verfahren.

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History of Science

Three traditions



Administration

- money
- (felt) urgency
- outlines
- practices



Process control

- outlines
- technology (analogue)
- practices



Science

- concepts
- calculations
- outlines

History of Science

Three traditions



Administration

- money
- (felt) urgency
- **outlines**
- practices



Process control

- **outlines**
- technology (analogue)
- practices



Science

- concepts
- calculations
- **outlines**

History of Science

Three traditions



Administration

- **money**
- (felt) urgency
- **outlines**
- practices



Process control

- **outlines**
- technology (analogue)
- practices



Science

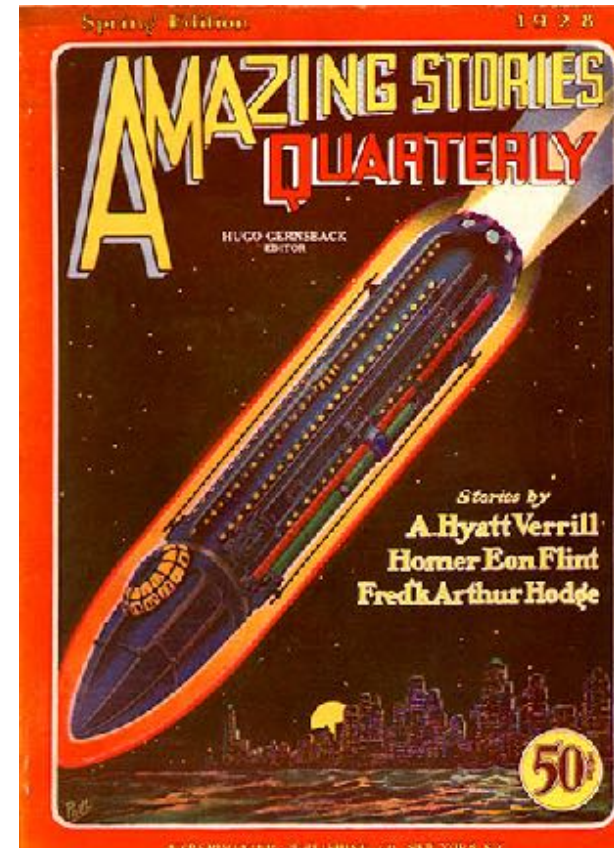
- concepts
- calculations
- **outlines**

History of Science

A prehistory of AI

- Human thinking and behaviour
- Language!

(these two lines would end up in cybernetics after World War II)



History of Science

Human thinking and behaviour

René Descartes (1596-1650): “Je pense, donc je suis”



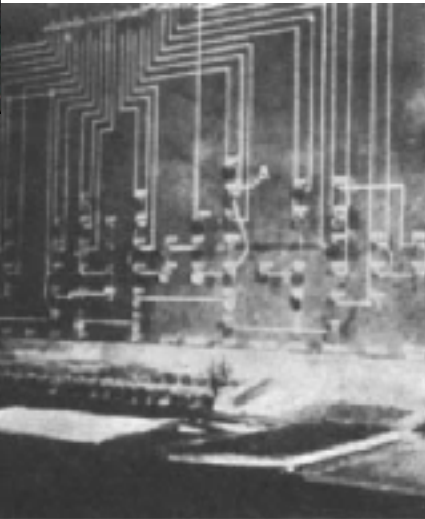
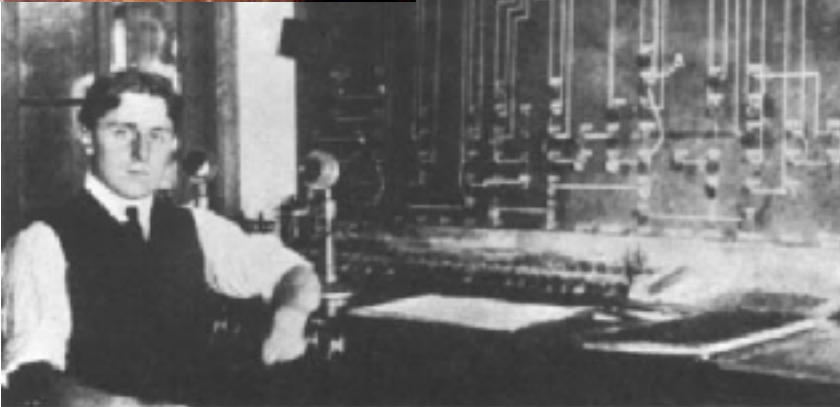
History of Science

Human thinking (and behaviour)



History of Science

Human (thinking and) behaviour



History of Science

Human thinking (and behaviour)

Teaching. Rise of teaching training and didactics coincides with the rise of the nation state.

How do we learn? Do we learn to think?

William Jevons (1835-1882)



History of Science

Human thinking and behaviour

Leonardo Torres Quevedo (1858-1936), *chess endgame* (1912)



History of Science

With hindsight

In literature and popular culture, artificially created “humans” pop up long before suggestions of automatic computers.

Mary Shelly, *Frankenstein* (1818)

Karel Čapek, *RUR* (1920), coining “robot”

Fritz Lang, *Metropolis* (1927)

cf. Charlie Chaplin, *Modern times* (1936)

BUT: does AI exist without computers?



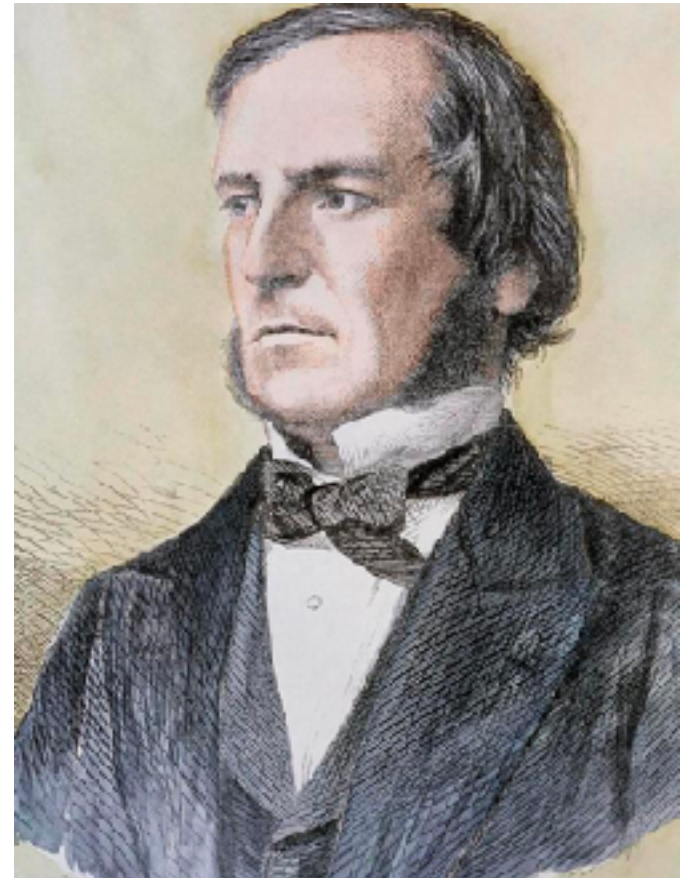
History of Science

Language! (human thinking)

Boolean logic: algebra, language of calculating with truth and false.

1854: *An investigation of the laws of thought*

Expressing thought in formulae
and being able to calculate with
these!



History of Science

Language!

Gotlob Frege (1848-1925), *Begriffsschrift* (1879)

Remained completely obscure until the early 20th c

Bertrand Russell (1872-1970) and
Alfred Whitehead (1861-1947),
Principia Mathematica
(1910-1913)

1	$\forall y \neg P(y)$	
2	$\exists x P(x)$	
3	u $P(u)$	
4	$\forall y \neg P(y)$	R, 1
5	$\neg P(u)$	$\forall E$, 4
6	$\perp$	$\neg E$, 3,5
7	$\perp$	$\exists E$, 2, 3-6
8	$\neg \exists x P(x)$	$\neg I$, 2-7

History of Science

Language!

1928 David Hilbert (1862-1943)

Entscheidungsproblem (Decision problem): is there a way to calculate the (un)truth of a given expression in a given mathematical system (with fixed “language”?)

1931 Kurt Gödel (1906-1978): no way!

History of Science

Language!

Alonzo Church (1903-1995), lambda calculus —>
1936: *Journal of symbolic logic*

Alan Turing (1912-1954), On computable numbers
(introducing the a-machine)



History of Science

Language!

Learning a language by describing it: phonology, morphology, syntax, vocabulary

VOWEL CHART AND SYMBOLS

Monophthongs

Front.		Back	
i:		u:	High
ɪ		ʊ	
e	ə	ɜ:	Mid
æ	ʌ	ɔ:	Low mid
ʌ		ɑ:	Low

heart /hɜ:t/, *hit* /hɪt/, *het* /het/, *hat* /hæt/, *hard* /hɑ:d/, *ho(stom)* /hɒtəm/, *hull* /hʊl/,
but /bʌt/, *bought* /bɔ:t/, *boor* /bʊə/, (*brother* /brʌðə/, *bird* /bɜ:d/

<i>Diphthongs</i>	rising:	ai, au, oi	<i>bite</i> /baɪ/, <i>how</i> /haʊ/, <i>boil</i> /boɪ/
		ei, æi	<i>hair</i> /heɪ/, <i>here</i> /hɛə/
	centring:	ie, ee, oo	<i>pie</i> /praɪ/, <i>pear</i> /peə/, <i>poor</i> /pʊə/

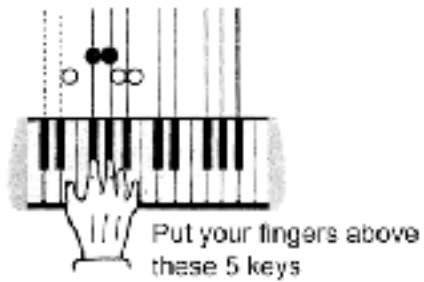
History of Science

Language!

International artificial language: Volapük (early 1880s), Esperanto (1887), Ido (1894)...

From within the peace movement (lady Victoria Welby - signífics)



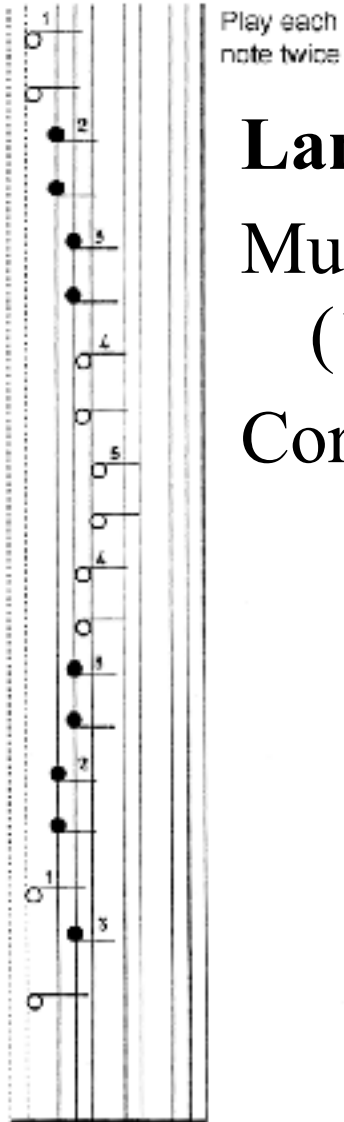


History of Science

Language!

Music learning: klavarscribo
(1931)

Cornelis Pot (1885-1977)



History of Science

THE END (for now)

Do not forget to:

- Enroll in a group
- Make the assignment
- Hand in the assignment
- Prepare for next lecture (see Canvas! Use the quiz if you think that helps)
- Use the discussion board or mail me with any questions about book or lecture!

